

```
[1] 1
attr(,"response")
[1] 1
attr(,".Environment")
<environment: R_GlobalEnv>
attr(,"predvars")
list(y, x)
attr(,"dataClasses")
      y      x
"numeric" "numeric"
```

```
> r$df.residual
[1] 3
```

```
> r$qr
$qr
      (Intercept)      x
1 -2.2360680 -11.1803399
2  0.4472136  6.3245553
3  0.4472136 -0.1954395
4  0.4472136 -0.5116673
5  0.4472136 -0.8278950
attr(,"assign")
[1] 0 1
```

```
$qraux
[1] 1.447214 1.120788
```

```
$pivot
[1] 1 2
```

```
$tol
[1] 1e-07
```

```
$rank
[1] 2
```

```
attr(,"class")
[1] "qr"
```

```
> r$xlevels
named list()
```

The `confint()` function computes the confidence intervals for the parameters in the fitted model. For example:

```
> confint(r)
      2.5 % 97.5 %
(Intercept)      1      1
x              1      1
```

The `confint()` function accepts the following arguments:

Argument	Description
object	fitted model object
parm	vector of numbers or names for the parameters
level	confidence level required
...	additional arguments

The analysis of variance tables for the fitted model can be obtained using the `anova()` function, as follows:

```
> anova(r)
Analysis of Variance Table
```

```
Response: y
      Df Sum Sq Mean Sq    F value    Pr(>F)    
x         1      40      40 1.2169e+32 < 2.2e-16 ***
Residuals  3        0        0
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Consider the `mtcars` data set available in the `lattice` library:

```
> head(mtcars)
      mpg cyl disp  hp drat   wt  qsec vs am gear carb
Mazda RX4  21.0   6  160 110 3.90 2.620 16.46  0  1   4   4
Mazda RX4 Wag  21.0   6  160 110 3.90 2.875 17.02  0  1   4   4
Datsun 710   22.8   4  108  93 3.85 2.320 18.61  1  1   4   1
Hornet 4 Drive  21.4   6  258 110 3.08 3.215 19.44  1  0   3   1
Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0   3   2
Valiant     18.1   6  225 105 2.76 3.460 20.22  1  0   3   1
```

We can fit a regression between the number of cylinders and horsepower using the `lm()` function, as follows:

```
> m <- lm(cyl~hp, data=mtcars)
> print(m)
```

```
Call:
lm(formula = cyl ~ hp, data = mtcars)
```

```
Coefficients:
(Intercept)      hp
   3.00680    0.02168
```

A summary of the fitted model is shown below:

```
> summary(m)
```

```
Call:
lm(formula = cyl ~ hp, data = mtcars)
```